

Ex no: 5(b)	Managing and Operating Docker Containers: Start, Tag, Rename and File Transfers
Date:	

Aim:

The aim of this project is to understand and demonstrate the usage of Docker by managing and operating containers. This includes starting and accessing existing containers, tagging and renaming Docker images, transferring files between the host and containers, and verifying container execution. The project emphasizes optimizing containerized applications deployment, portability, and scalability.

Docker:

Docker is an open-source containerization platform that automates the process of creating, deploying, and managing containers. It enables developers to package an application along with all its dependencies into a container image, ensuring the application runs consistently across different environments. This eliminates the common problem of "it works on my machine," as the container behaves the same way whether it is running on a laptop, server, or cloud.

Purpose of Docker:

The purpose of this project is to develop practical skills in managing Docker containers and images. It focuses on starting and operating containers, tagging and renaming images, transferring files between the host and containers, and ensuring efficient deployment and portability of applications in a containerized environment.

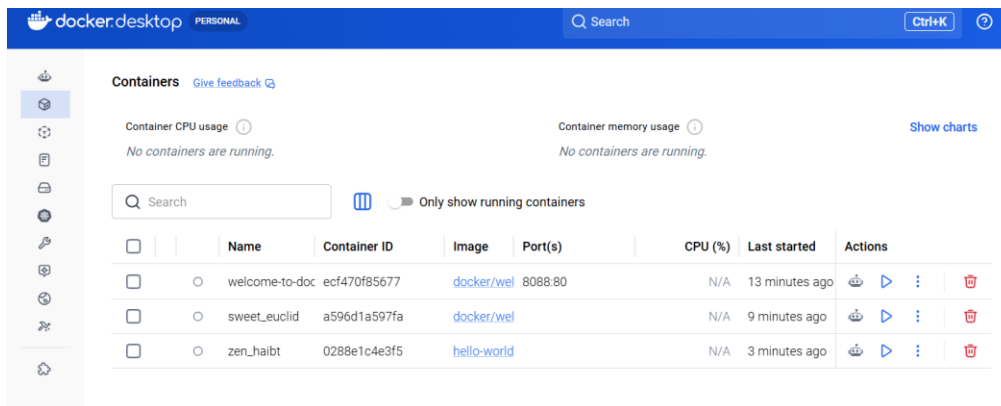
Commands and Output:**Step 1: Start Node Container**

docker start <container_name>

```
C:\Users\Reena>docker start unruffled_franklin
unruffled_franklin
```

docker exec -it <container_name> node

```
C:\Users\Reena>docker run -it node
Welcome to Node.js v26.7.0.
Type ".help" for more information.
> let a=1
undefined
> let b =2
undefined
> console.log(a+b)
3
undefined
```

**OUTPUT:****Step 2: Enter the Docker Container Using Bash**

```
docker exec -it <docker_name> /bin/bash
```

```
root@15f98f8f5df1:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@15f98f8f5df1:/# cd /usr/local/bin
root@15f98f8f5df1:/usr/local/bin# ls
docker-entrypoint.sh node nodejs npm npx
root@15f98f8f5df1:/usr/local/bin# node
Welcome to Node.js v26.7.0.
Type ".help" for more information.
>
```

Step 3: Rename the Docker Container (New cmd)

```
docker rename <current_container_name> <new_container_name>
```

```
C:\Program Files\Docker\Docker\resources\bin>docker rename node_image my_node_image
C:\Program Files\Docker\Docker\resources\bin>
```

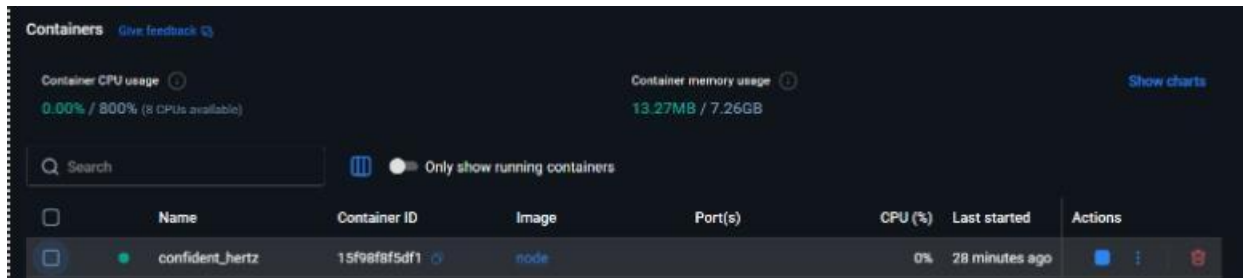
OUTPUT:

	my_node_image	dc8490f78a9a	node	0%	2 minutes ago				
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Step 4: Pull a Docker Image

```
docker pull hello-world
```

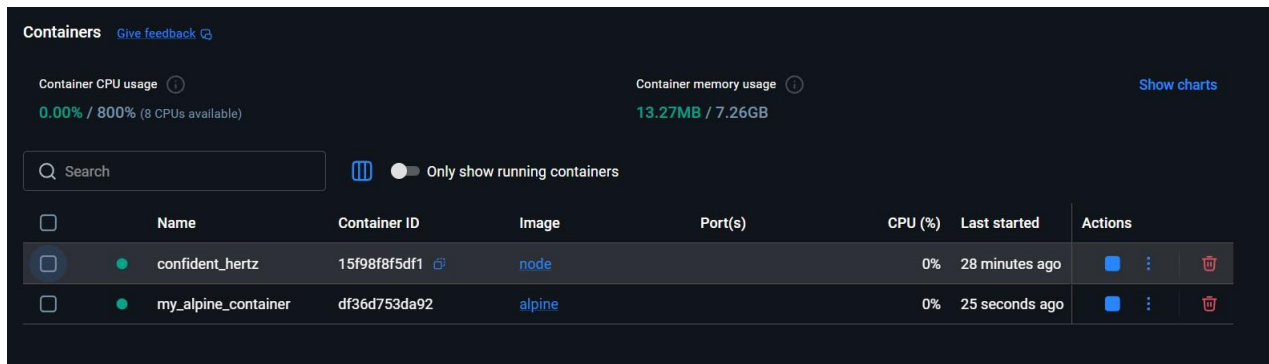
```
C:\Users\Reena>docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
```

OUTPUT:**Step 5: Run Alpine Container and Explore File System**

(Now go back to the previous cmd prompt and give this)

docker run -it alpine

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker run -it --name my_alpine_container alpine / #
```

OUTPUT:**Step 6: Copy a Local File into the Docker Container**

docker cp "C:\Users\<your-path>\my-example.txt" my_alpine:/example.txt

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker cp .\my-example.txt my_alpine_container:/example.txt
Successfully copied 1.54kB to my_alpine_container:/example.txt
```

Step 7: Verify the Copied File Inside the Container

```
docker run -it <container_name> sh
```

```
cd /var
```

```
ls /
```

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker exec -it my_alpine /bin/sh
/ # ls
bin      etc      home     media    opt      root     sbin     sys      usr
dev      example.txt lib      mnt      proc     run      srv      tmp      var
/ #
```

Step 8: Copying a File from Docker Container to Host

```
docker cp <container_name>:<container_path> <host_path>
```

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker cp my_alpine:/example.txt ./example_copy.txt
Successfully copied 1.54kB to C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile\example_copy.txt
```

Name	Status	Date modified	Type	Size
 example_copy		19-08-2026 16:39	Text Document	0 KB
 my-example		19-08-2026 16:39	Text Document	0 KB

Problem Understanding & approach (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)

Result:

The project successfully demonstrates Docker operations including container creation, image, management, file transfer, and in-container verification, validating Docker's efficiency in scalable and portable application deployment.